

Paper III - Subspeciality of Urology (practice paper, MCh Urology Summer 2015 set)

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III-1. Discuss various ureteric stents. Discuss ureteroscopy with or without stenting.

Type: LAQ (25 marks)

Introduction

- A stent keeps the upper tract draining past obstruction, injury or an anastomosis. **Zimskind 1967** placed the first indwelling splint; **Finney 1978** introduced the self-retaining **double-J (double-pigtail)** stent.
- Ideal stent: easy to insert, resists migration, drains, radio-opaque, resists encrustation, few symptoms.

Types and materials

Feature	Details
Size	3–8.5 Fr (adult usually 6 Fr); adults 20–30 cm by height, children about age + 10 cm
Polymers	Polyurethane, silicone (least encrusting, soft); hydrocoated silicone causes less pain
Metallic (Resonance)	Resists extrinsic compression: malignant obstruction, retroperitoneal fibrosis
Others	Hybrid metal-reinforced, thermo-expandable nitinol (Memokath), biodegradable (experimental)
Designs	Single-pigtail diversion stent; open-ended catheter to identify the ureter; modified bladder ends (Polaris loop: no benefit)

Also know: anatomy: three narrowings (PUJ (pelviureteric junction), iliac vessels, intramural ureter, the narrowest at 3–4 mm). Placement: retrograde pyelogram first, stent over a wire under fluoroscopy; antegrade route if retrograde fails.

Diagram to draw: double-J stent in situ with both coils and side holes.

Indications

- Obstruction: infected obstructed kidney (stent equals PCN (percutaneous nephrostomy)), malignancy, stricture, pregnancy.
- After URS (ureteroscopy), pyeloplasty, reimplantation, transplant; injury, leak, fistula; pre-stenting a tight ureter.

Stent symptoms and complications

- About **4 in 5** have urgency, frequency, haematuria or flank pain, measured by the **USSQ (Ureteral Stent Symptom Questionnaire)**. Cause: **intravesical material irritating the bladder and reflux up the stent**. Diameter and length matter little.
- Relief: **alpha-blockers and antimuscarinics**; intraureteral placement; shortest dwell; opioid-free ERAS (enhanced recovery after surgery) analgesia.
- **Misplacement** (submucosal) and **migration** (too short).
- **Stent failure** from compression, debris or encrustation: worst with malignant or irradiated ureter, bilateral obstruction; then PCN.
- **Encrustation** rises with dwell time, infection and stone-forming urine. **Forgotten stent** causes sepsis and renal loss; keep a register. **FECal (Forgotten, Encrusted, Calcified) grade I-V** by coil involvement.
- **Encrusted stent**: never pull by force; NCCT (non-contrast computed tomography); clear the **bladder coil**, then **ureter**, then **renal coil** (flexible URS or PCNL (percutaneous nephrolithotomy)).

Ureteroscopy

	Semirigid	Flexible
Tip	7 Fr or less	4.5–7.5 Fr; 3.6 Fr channel
Use	Distal and mid ureter	Upper ureter, kidney
Deflection	None	Over 270°

- **Safety guidewire** first (Segura's dictum). Tight ureter: stent and return after **2-4 weeks**; routine pre-stenting is unnecessary.
- **UAS (ureteral access sheath)**, usually 12/14 Fr: lowers intrarenal pressure and eases repeated passes, but **severe injury in about 1 in 8**; less after pre-stenting.
- **Ho:YAG (holmium:yttrium-aluminium-garnet)** or **TFL (thulium fibre laser)**: dusting (low energy, high frequency) or fragment and basket. Keep pressure and fluid temperature low with outflow.
- **Also know**: anti-retropulsion devices; digital and single-use scopes; URS is best for patients who cannot stop anticoagulants; one antibiotic dose unless urosepsis.
- Complications: sepsis about 5% (treat positive urine culture first), stricture about 3%, bleeding under 1%, perforation (stop and stent), avulsion under 1% (PCN, delayed repair).

Stent after URS?

- **Denstedt 2001** randomised trial and a Cochrane review: URS is safe without a stent.
- **AUA (American Urological Association)**: stent may be omitted when anatomy and renal function are normal, there is no ureteric injury and no second stage is planned. **EAU (European Association of Urology)** agrees.
- **Stent needed**: perforation, stricture or tight ureter, infection, planned second stage, solitary kidney, impacted stone.
- Removal at 3 days gave more unplanned visits than at 7 days; a string allows home removal. Stenting at SWL (shock wave lithotripsy) does not improve stone passage.

Uncomplicated URS	-> no stent (or short string stent)
Complicated	-> double-J: injury 2-6 wk; second stage until re-look

Recent advances

- Suction sheaths, intrarenal pressure sensors, biodegradable stents.

Key points to remember:

- Finney 1978 double-J; symptoms from intravesical stent and reflux.
 - Encrusted stent: bladder coil, ureter, then kidney.
 - Denstedt 2001: no routine stent after uncomplicated URS.
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III-2. Describe medical management of VUR. Describe treatment of an 8-yr male child with b/l grade IV VUR.

Type: LAQ (25 marks)

Definition and pathophysiology

- **VUR (vesicoureteral reflux)**: retrograde urine flow to the upper tract. **Primary**: deficient UVJ (ureterovesical junction) flap-valve; tunnel length to diameter should be about **5:1** (Paquin). An early ureteric bud gives a lateral orifice, short tunnel and often dysplasia.
- **Secondary**: high bladder pressure from PUV (posterior urethral valves), neurogenic bladder or BBD (bladder and bowel dysfunction).
- In about **30% of children with UTI (urinary tract infection)** and 30% of siblings; commoner in girls, infant boys, white children.
- **Infection, not reflux, injures the kidney**. Infected urine plus intrarenal reflux causes scars, hypertension and CKD (chronic kidney disease); some high-grade "scars" are congenital dysplasia.

International grading (VCUG, voiding cystourethrogram)

Grade	Findings
I-II	Ureter only; up to calyces without dilatation
III	Mild-moderate dilatation, minimal forniceal blunting
IV	Moderate dilatation and tortuosity; papillary impressions kept
V	Gross dilatation; papillary impressions lost

- **Also know**: grading varies between observers; UDR (ureteral diameter ratio) predicts resolution better.

Diagram to draw: grades I-V and a Cohen cross-trigonal reimplant.

Natural history

- Resolution falls with grade and age: grade II 63-85%, grade III about 50%, IV-V rarely. **Bilateral grade IV persists most** (AUA (American Urological Association) 1997: about 10% resolve in 5 years).

Evaluation

- History of febrile UTIs and BBD; BP (blood pressure), spine, urine protein, creatinine. **RBUS (renal-bladder ultrasound)**: a normal scan does not exclude VUR.
- **VCUG**: grade and urethra. **Nuclear cystogram** (lower dose) for follow-up; **ceVUS (contrast-enhanced voiding urosonography)** avoids radiation.
- **DMSA (dimercaptosuccinic acid)**: scars and split function; a defect persisting 6 months is a scar.

Medical management

- **Aim**: prevent febrile UTI and scars while VUR resolves. **High risk**: female, infant, bilateral, grade III-V, BBD, scars, multiple febrile UTIs.
- **CAP (continuous antibiotic prophylaxis)**: one nightly dose, about a quarter of the treatment dose.

Drug	Dose	Note
TMP-SMX (trimethoprim-sulfamethoxazole)	2 mg/kg TMP	Avoid under 2 months (kernicterus)

Drug	Dose	Note
Nitrofurantoin	1-2 mg/kg	Avoid in G6PD (glucose-6-phosphate dehydrogenase) deficiency; rare lung toxicity
Trimethoprim or amoxicillin	—	Young infants

- **RIVUR (2014)**, 607 children: recurrent UTI **23.6% to 12.9%** on TMP-SMX; **scarring unchanged**; most benefit with BBD.
- **Swedish Reflux Trial** (grade III-IV, age 1-2): febrile UTI in girls 19% on CAP, 23% after endoscopy, 57% on surveillance; **no difference in boys**.
- **Also know:** IRSC (International Reflux Study in Children) and Birmingham trials: surgery and medical care gave equal scarring; PRIVENT: modest 6% absolute UTI reduction.
- **Treat BBD** (timed voiding, constipation, anticholinergics): it raises breakthrough UTI and lowers resolution and endoscopic success.
- **Circumcision:** about 4 boys with VUR circumcised prevent one UTI.
- After stopping CAP, 8-10% have a febrile UTI.
- **Intervene for** breakthrough febrile UTI, new scars, persistent high grade, non-adherence.

Case: 8-year-old boy, bilateral grade IV

- Exclude PUV, neurogenic bladder and BBD; DMSA, creatinine, BP.
- This rarely resolves at 8 years, but older boys have low UTI risk, so treating an **asymptomatic** boy is **controversial**.
- **No UTI, no scars:** observation or CAP with BBD therapy; correct if febrile UTI or new scar.
- **Breakthrough UTI, scars or falling function:** correct surgically.

Option	Key facts
Endoscopic Dx/HA (dextranomer/hyaluronic acid)	STING (subureteric Teflon injection) or HIT (hydrodistension implantation); "volcano" mound; about 60% per injection in grade IV; 20% recurred by 2 years; late obstruction reported
Cohen cross-trigonal	Commonest, over 98% success; one bladder opening treats both sides; later ureteroscopy difficult
Lich-Gregoir (extravesical)	No haematuria or spasm; retention up to 10% after bilateral repair
RALUR (robot-assisted laparoscopic ureteral reimplantation)	Less pain; results approaching open

- Open reimplantation succeeds in **95.9%** (Elder 1997); others: Politano-Leadbetter, Glenn-Anderson. **Choice here: bilateral Cohen.**
- After surgery: ultrasound at 6-12 weeks. Complications: transient obstruction, contralateral VUR (about 1 in 10, usually resolves).

Exclude PUV/neurogenic bladder, treat BBD
 -> febrile UTI, scars or renal loss? Yes -> bilateral Cohen (or Dx/HA)
 No -> observe or CAP; correct if UTI/scar

Key points to remember:

- Grade V loses papillary impressions; bilateral grade IV rarely resolves.
- RIVUR: CAP halves recurrent UTI but not scarring.
- Open reimplantation 95.9%; bilateral extravesical repair risks retention.

III-3a. Sacral neuromodulation

Type: SAQ (10 marks)

Definition and mechanism

- **SNM (sacral neuromodulation)**, introduced by Tanagho and Schmidt: continuous low-amplitude **S3** stimulation through a tined lead and IPG (implantable pulse generator).
- It acts on **afferents**, which modulate forebrain awareness. In OAB (overactive bladder) this damps the PMC (pontine micturition centre); in retention it restores filling sensation, so the PMC switches on and the sphincter relaxes. Hence it treats **both**.

Indications (third line)

FDA (Food and Drug Administration) approved	Not approved
Refractory UUI (urgency urinary incontinence) 1997; urgency-frequency 1999	Bladder pain, pelvic pain
Non-obstructive retention (e.g. Fowler's syndrome) 1999; faecal incontinence 2011	Neurogenic bladder

- Retention needs some detrusor function; success falls with age.

Technique

- **Test: PNE (percutaneous nerve evaluation)** 5–7 days, or **staged tined lead** up to 4 weeks (more reliable; used for retention).
- S3: about **9 cm above the coccyx tip, 1–2 cm lateral**. Response: **perineal bellows plus great-toe flexion**.
- **Implant if at least 50% improvement** in a diary parameter (or catheterisations); IPG in the buttock.
- **Also know:** 10–16 Hz at sensory threshold; bilateral is not better.

Outcomes

- **InSite:** 61% vs 42% success over medical therapy at 6 months.
- **ROSETTA:** onabotulinumtoxinA 200 U slightly better for UUI but more UTI (urinary tract infection) (24% vs 10%) and catheterisation.

Complications

- Pain, lead migration, loss of effect, infection; **revision over 30% at 5 years**; costliest option.

Recent advances

- Rechargeable or 10-year batteries; full-body MRI (magnetic resonance imaging) conditional. Alternatives: PTNS (percutaneous tibial nerve stimulation), pudendal stimulation.

Diagram to draw: lead in S3, IPG in the buttock.

Key points to remember:

- Afferent modulation treats OAB and retention.
- S3: bellows plus great-toe flexion.
- Implant if test improves at least 50%.

III-3b. Uses of buccal mucosa in urology

Type: SAQ (10 marks)

Introduction

- **BMG (buccal mucosal graft)** is the workhorse graft for urethral reconstruction. **Humby 1941** (hypospadias); **Bürger 1992** and **El-Kasaby 1993** (stricture).

Why it works

- Thick, **non-keratinised**, hairless epithelium used to a wet environment.
- **Thin, highly vascular lamina propria**: rapid imbibition and inosculation, so take is excellent. Little contraction; low donor morbidity.
- A graft lives on its bed, so it does best on thick vascular beds (bulb, corpora).

Harvest

- Nasotracheal intubation; about 4–6 × 2–2.5 cm per cheek, **1 cm from the commissure, avoiding Stensen's duct** (opposite the upper second molar).
- Defat, cut about 10% oversize, fenestrate, quilt to the bed. Alternatives: **lingual, lip** (thinner, for the glans).
- Donor problems: pain, numbness, mouth tightness.

Uses

Area	Use
Bulbar stricture	Dorsal onlay (Barbagli), ventral onlay (needs healthy spongiosum), Asopa inlay, augmented anastomosis
Panurethral	Kulkarni one-stage dorsal onlay
Staged (Johanson, Bracka)	Lichen sclerosus , failed hypospadias; tubularise after 6 months or more
Hypospadias	Redo repairs; dorsal inlay for a narrow plate
Others	BMG perineal urethrostomy, ureteroplasty, bladder neck contracture, female urethra

Outcomes

- Bulbar BMG urethroplasty about 85–90%; dorsal equals ventral; graft equals flap; onlay beats tube.
- Results fall with time and in penile, redo and lichen sclerosus cases.
- **Avoid skin in lichen sclerosus**; it recurs in skin (rarely even in buccal mucosa).
- **Also know**: other grafts: full-thickness skin, bladder mucosa (meatal prolapse), meshed split skin (staged only).

Recent advances

- Tissue-engineered oral mucosa (investigational).

Diagram to draw: bulbar cross-section with dorsal onlay, ventral onlay and Asopa inlay.

Key points to remember:

- Thin vascular lamina propria gives excellent take.
- Staged BMG for lichen sclerosus.
- Bulbar success about 85–90%.

III-3c. HSG in neonates

Type: SAQ (10 marks)

HSG (hysterosalpingography) is not done in neonates, so this is taken as "USG (ultrasonography) in neonates".

Introduction

- First-line, radiation-free imaging, mainly for antenatal **UTD (urinary tract dilation)** (1–3% of pregnancies).

Timing and technique

- First scan after 48 hours:** newborns are relatively dehydrated, so earlier scans under-read dilatation.
- Scan early if outlet obstruction is suspected;** catheterise and give prophylaxis.
- Normal: **prominent hypoechoic pyramids** (not hydronephrosis); **APD (anteroposterior pelvic diameter)** under 10 mm.

Grading

- SFU (Society for Fetal Urology) 0–4:** pelvis; major calyces; minor calyces; plus parenchymal thinning.
- UTD 2014** combines both (prenatal A1, A2–3; postnatal P1–P3):

Grade	Findings	Plan
P1	APD 10 to <15 mm or central calyces	USG at 3–6 months; no VCUG (voiding cystourethrogram) or prophylaxis
P2	APD 15 mm or more, peripheral calyces or dilated ureter	VCUG, prophylaxis optional (give if ureter >7 mm)
P3	Abnormal parenchyma or bladder	USG at 1 month, VCUG, prophylaxis, MAG3 (mercaptoacetyltriglycine) scan

Specific findings

Condition	USG
PUJO (pelviureteric junction obstruction)	Dilated pelvis, normal ureter; pyramids under 3 mm thick predict pyeloplasty
VUR (vesicoureteral reflux)	Up to 25% have a normal first scan
PUV (posterior urethral valves)	Bilateral hydroureteronephrosis, thick bladder, dilated posterior urethra (keyhole sign); VCUG before discharge
Ureterocele	Thin-walled intravesical cyst; dilated upper pole
MCDK (multicystic dysplastic kidney)	Non-communicating cysts, no central pelvis

- Also know:** ARPKD (autosomal recessive polycystic kidney disease): large bright kidneys; mesoblastic nephroma; renal vein thrombosis; adrenal haemorrhage; twinkling artefact (stones).

Recent advances

- ceVUS (contrast-enhanced voiding urosonography).

Diagram to draw: SFU grades 0–4.

Key points to remember:

- First scan after 48 hours, except suspected PUV.
 - UTD: P1 APD 10 to <15 mm; P2 15 mm or more; P3 abnormal parenchyma or bladder.
 - Keyhole sign suggests PUV.
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III-3d. Types of penile prosthesis. What are biofilms in penile prosthesis?

Type: SAQ (10 marks)

Introduction

- For ED (erectile dysfunction) failing less invasive treatment; satisfaction **80-90%**. Repairs: about 15% at 5 years, 30% at 10.

Types

Type	Examples	Features
Malleable	Tactra (nitinol), Genesis	Simple, cheap, no mechanical failure; always rigid
2-piece inflatable	Ambicor (discontinued)	No reservoir; poor flaccidity
3-piece inflatable	AMS 700, Titan	Best rigidity and true flaccidity ; cylinders, scrotal pump, reservoir

- Penoscrotal or infrapubic approach. Reservoir in Retzius, or **ectopic submuscular** after pelvic surgery (lockout valve prevents autoinflation).
- **Also know:** erosion, pump migration, glans hypermobility, cylinder aneurysm; Peyronie's disease may need modelling.

Biofilm

- **Definition:** a structured community of microorganisms in a self-made **polysaccharide matrix**, stuck to the implant.
- **Stages:** attachment, microcolonies, maturation, dispersal (clinical infection).
- **Organisms:** coagulase-negative staphylococci commonest; also *S. aureus*, Gram-negatives, *Candida*.
- **Why antibiotics fail:** poor penetration and dormant bacteria, which survive up to **1000-1500 times** normal killing levels.
- **Relevance:** organisms from the first operation lie dormant; revising an uninfected device gives about 10% infection, cut to about 3% by **antiseptic washout**.

Prevention (infection 1-3%)

- Glucose under 200 mg/dL on the day; treat skin and urine; MRSA (methicillin-resistant *S. aureus*) decolonisation if carrier.
- **Chlorhexidine-alcohol** skin preparation; vancomycin or cephalosporin plus gentamicin for up to 24 hours.
- **Coatings halve infection:** InhibiZone (rifampin-minocycline); hydrophilic coating dipped in antibiotic.
- **No-touch technique:** under 0.5%.

Infected implant

- Pus, fixed pump or exposed parts: remove **all** components.
- **Mulcahy salvage:** explant, antiseptic washes, immediate new device; **84%** success. Fails with early onset, aggressive organisms, cellulitis.

Diagram to draw: 3-piece device; biofilm life cycle.

Key points to remember:

- 3-piece inflatable gives best rigidity and flaccidity.
 - Biofilm: microbes in a matrix; antibiotics fail.
 - Infection 1-3%; Mulcahy salvage 84%.
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III-3e. Uretero-vaginal fistula

Type: SAQ (10 marks)

Definition and aetiology

- **UVF (uretero-vaginal fistula):** ureter-vagina communication causing continuous leakage.
- Usually **unrecognised distal ureteric injury**; about **three-quarters follow abdominal hysterectomy**, often uneventful (ureteric injury in 0.2–2.4%). Others: obstetric, cancer, radiation.
- Danger points: pelvic brim, **under the uterine artery**, beside the uterosacral ligament, vaginal fornix. Mechanisms: ligation, crush, thermal, ischaemia.

Clinical features

- **Continuous leakage with normal voiding:** the other kidney fills the bladder.
- Appears **within 4 weeks**, often with fever or flank pain, as thermal or ischaemic injury necroses.

	VVF (vesicovaginal fistula)	UVF
Voiding	Reduced	Normal
Upper tract	Normal	Hydronephrosis common
First line	Catheter, repair	Stent

Investigations

- **Double dye test:** oral phenazopyridine plus blue dye in the bladder; **orange tampon = UVF**, blue = VVF. Three-swab test: wet unstained swab.
- **CT (computed tomography) urography**, delayed phase (MR (magnetic resonance) urography if contrast is contraindicated).
- **Cystoscopy and retrograde pyelogram**, with a stent at the same sitting.
- **Exclude coexisting VVF** (up to 25% of VVF have UVF) and bilateral injury.

Management

- **Stent first:** retrograde double-J if a wire passes; otherwise PCN (percutaneous nephrostomy) and antegrade stent. **Stenting within 2 weeks** works best; remove at **4–6 weeks** and re-image.
- **Surgery** if stenting fails: **ureteroneocystostomy** with **psoas hitch** or **Boari flap** (up to 15 cm); downward nephropexy or ileal ureter for longer gaps; open, laparoscopic, robotic or vaginal. Early repair equals delayed. Success **90% or more**.
- Follow-up ultrasound: stricture may appear up to 30 months later.

Prevention

- Recognise injury intraoperatively (cystoscopy); prophylactic stents only if high risk.

Diagram to draw: double dye test; psoas hitch.

Key points to remember:

- Continuous leak with normal voiding; orange tampon = UVF.
- Exclude coexisting VVF.
- Early stent; reimplant with psoas hitch if it fails.

III-3f. Prune belly syndrome

Type: SAQ (10 marks)

Definition

- **PBS (prune belly syndrome)**, Eagle-Barrett or triad syndrome: **deficient abdominal wall muscles, dilated poorly contracting urinary tract, bilateral intra-abdominal testes.**
- 1 in 29,000–44,000 births; **95% male. Renal dysplasia, not the abdominal wall, decides survival.**

Pathogenesis (unproven)

- Fetal **urethral obstruction**, though most have none after birth; **lateral plate mesoderm defect**; intrinsic ureteric defect; yolk sac defect. Mostly sporadic; smooth-muscle genes (MYOCD, FLNA) in families.

Features

- **Kidneys**: dysplasia in about 50%. **Ureters**: tortuous, worse distally; collagen replaces muscle, so peristalsis fails; **VUR (vesicoureteral reflux) about 75%.**
- **Bladder**: large, smooth, urachal diverticulum (urachus patent in 25–30%). **Prostatic hypoplasia**, dilated posterior urethra; atresia or megalourethra.
- **Testes**: intra-abdominal; azoospermia usual.
- **Extra-genitourinary (about 75%)**: pulmonary hypoplasia; cardiac 10%; gastrointestinal at least 30%; orthopaedic up to 65%.

Woodard classification

	Features	Outlook
I	Oligohydramnios, pulmonary hypoplasia , dysplasia	Perinatal death
II	Full triad, moderate uropathy	Risk of CKD (chronic kidney disease); debated
III	Mild, stable function	Good; majority

- **Also know**: RUBACE severity score (renal, ureter, bladder, abdominal wall, cryptorchidism, extra-genitourinary); "pseudo-prune" presents late.

Management

- **Neonatal**: lungs and heart first; creatinine trend, ultrasound, DMSA (dimercaptosuccinic acid) at 4–6 weeks, prophylaxis, circumcision. VCUG (voiding cystourethrogram) only if renal insufficiency or obstruction.
- Category III: surveillance. Sepsis or rising creatinine: wide **cutaneous vesicostomy.**
- **Reconstruction** after 3–6 months: ureteric tailoring and reimplantation using the healthier proximal ureter; reduction cystoplasty gives little lasting benefit.
- **Orchidopexy**: bilateral **transabdominal at about 6 months**; staged Fowler-Stephens later.
- **Abdominoplasty**: Randolph, Ehrlich, Monfort.

Prognosis

- 20% die perinatally; **up to 30% develop chronic renal failure**; nadir creatinine **under 0.7 mg/dL** is favourable.

Diagram to draw: VCUG showing large bladder and dilated posterior urethra.

Key points to remember:

- Triad; 95% male; dysplasia decides prognosis.
 - Woodard I-III.
 - Vesicostomy for drainage; orchidopexy at about 6 months.
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Rapid Revision Sheet - Paper III

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

III-1. Discuss various ureteric stents. Discuss ureteroscopy with or without stenting. (LAQ)

- Finney 1978 double-J; symptoms from intravesical stent and reflux.
- Encrusted stent: bladder coil, ureter, then kidney.
- Denstedt 2001: no routine stent after uncomplicated URS.

III-2. Describe medical management of VUR. Describe treatment of an 8-yr male child with b/l grade IV VUR. (LAQ)

- Grade V loses papillary impressions; bilateral grade IV rarely resolves.
- RIVUR: CAP halves recurrent UTI but not scarring.
- Open reimplantation 95.9%; bilateral extravesical repair risks retention.

III-3a. Sacral neuromodulation (SAQ)

- Afferent modulation treats OAB and retention.
- S3: bellows plus great-toe flexion.
- Implant if test improves at least 50%.

III-3b. Uses of buccal mucosa in urology (SAQ)

- Thin vascular lamina propria gives excellent take.
- Staged BMG for lichen sclerosus.
- Bulbar success about 85–90%.

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- First scan after 48 hours, except suspected PUV.
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- Early stent; reimplant with psoas hitch if it fails.

III-3f. Prune belly syndrome (SAQ)

- Triad; 95% male; dysplasia decides prognosis.
- Woodard I–III.
- Vesicostomy for drainage; orchidopexy at about 6 months.